MIL-I-21206B(WEP)
15 November 1960
Superseding
MIL-I-21206A(AER)
15 July 1958

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL; EFU-7/A (300-5000-15000 PPH)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers the design and performance requirements for the procurement of one type of a remote, single indicating, integrally lighted, fuel, rate of flow indicator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-451 - Glass, Flat and Corrugated, for Glazing, Mirrors and other Uses
QQ-P-416 - Plating, Cadmium (Electrodeposited)

Military

MIL-P-116 - Preservation, Method of
MIL-M-3171 - Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541 - Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872 - Soldering Process, General Specification for
MIL-I-7566 - Indicator System, Fuel Flowmeter, General Specification for
MIL-P-7936 - Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys

FSC 6620

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MIL-C-13335 - Coatings for Magnesium and Magnesium Alloys
MIL-S-20708 - Synchros, 60 and 400 Cycle, General Specification for
MIL-T-21205 - Transmitter, Rate of Flow, Fuel: Type TRU-12/A (300-5000-15,000 PPH)
MIL-L-25467 - Lighting, Integral, Instrument, General Specification for
MIL-C-26482 - Connectors, Electric, Circular, Miniature, Quick Disconnect

STANDARDS

Federal

FED STD No. 595 - Colors

Military

MIL-STD-704 - Electric Power, Aircraft, Characteristics and Utilization of
MS3134 - Connector, Electrical Receptacle, Solder Mounted
MS33558 - Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585 - Pointers, Dial, Standard Design of Aircraft Instrument
MS33639 - Case, Instrument, Clamp-Mounted, Aircraft

(When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention: Code CDS)

3. REQUIREMENTS

3.1 Qualification. - The indicator furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General specification. - The requirements of Specification MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

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3.3 Design and construction. -

3.3.1 General. - The indicator shall be designed to supply power and calibrated to operate satisfactorily with a 300-5000-15000 pph transmitter in accordance with Specification MIL-T-21205.

3.3.2 Calibration. - Each indicator dial shall be graduated in accordance with Figure 1. The calibration of fuel flow versus angular degrees shall be linear within ±0°15' of arc, except at the changeover point, and below 500 pph. A 25% deviation is permissible.

3.3.3 Remote indication. - Unless otherwise specified, the indicator shall be designed to indicate, at a remote station, the rate of flow in pounds per hour of aviation fuel when used in conjunction with a fuel flow transmitter in accordance with Specification MIL-T-21205.

3.4 Electrical. -

3.4.1 Power requirement. - The indicator shall operate from 115V ±20%, 400 cps ±20%, single phase ac power in accordance with the characteristics specified in Standard MIL-STD-704, except that the ratio of volts to frequency shall be 115/400 ±5%.

3.4.2 Power consumption. - The power consumption of the indicator shall not exceed 8.0 VA or 24 VA if used to supply power to one transmitter conforming to Specification MIL-T-21205.

3.4.3 Undervoltage protection. - The indicator shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within its correct range.

3.4.4 Electrical characteristics. - Sufficient electrical tests shall be performed by the manufacturer on the indicator and its components to insure satisfactory performance when connected in a complete flowmeter system as specified in 3.3.3.

3.4.4.1 Frequency inverter voltage. - The voltage supplied by the indicator to the transmitter shall not exceed 115V ± 30% 400 cps ±.25% when subjected to normal, high and low temperatures.

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3.4.5 Wiring. - The external wiring of the system shall be as shown in Figure 2. Internal wiring shall be neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be tied to the terminals prior to soldering, where practicable. The wiring shall be insulated from the indicator case except for the lighting circuit.

3.4.6 Electrical receptacle. - The indicator shall be provided with an electrical receptacle conforming to Specification MIL-C-26482 and to Standard MS3134-12P. The electrical receptacle shall be mounted on the rear surface of the case with the polarizing keyway of the receptacle at the top center of the receptacle shell.

3.5 Dial. -

3.5.1 Dial diameter. - The dial shall have a minimum diameter of 1 3/4 inches as measured at the outside ends of the scale graduations.

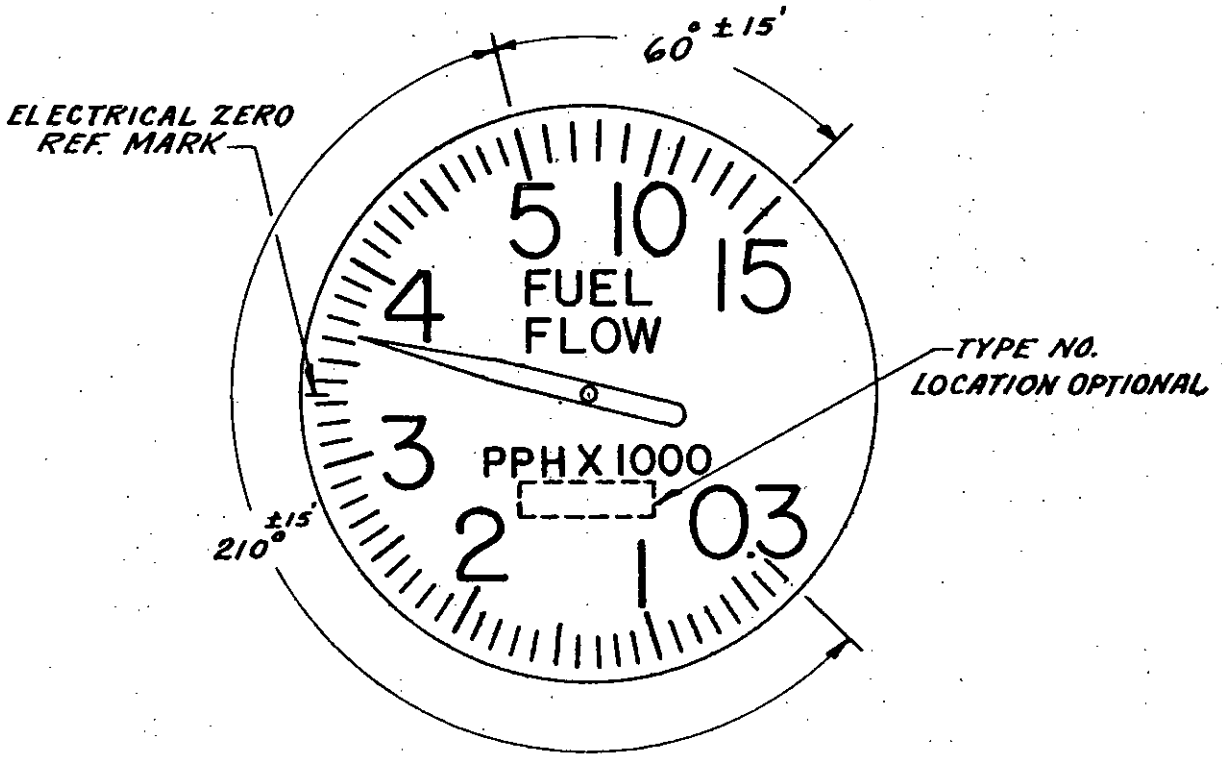
3.5.2 Dial markings. - Dial markings shall be durable and in accordance with Figure 1. The style and proportions of the letters and numerals placed on the dial shall conform to Standard MS33558. Numerals shall distinctly indicate the graduation to which each applies. The background of the dial shall be finished in a lusterless black color. The markings shall be finished in a lusterless white color. The dimensions of the markings shall be as listed in Table I.

Table I

DIAL MARKING

Marking	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Scale numerals 0.3 to 15	5/32	
Major graduations	5/32	0.031
Intermediate graduations	1/8	0.020
Minor graduations	3/32	0.020
Lettering "PPH X 1000"	3/32	
Lettering "Fuel Flow"	1/8	

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DIAL ARRANGEMENT

FIGURE 1

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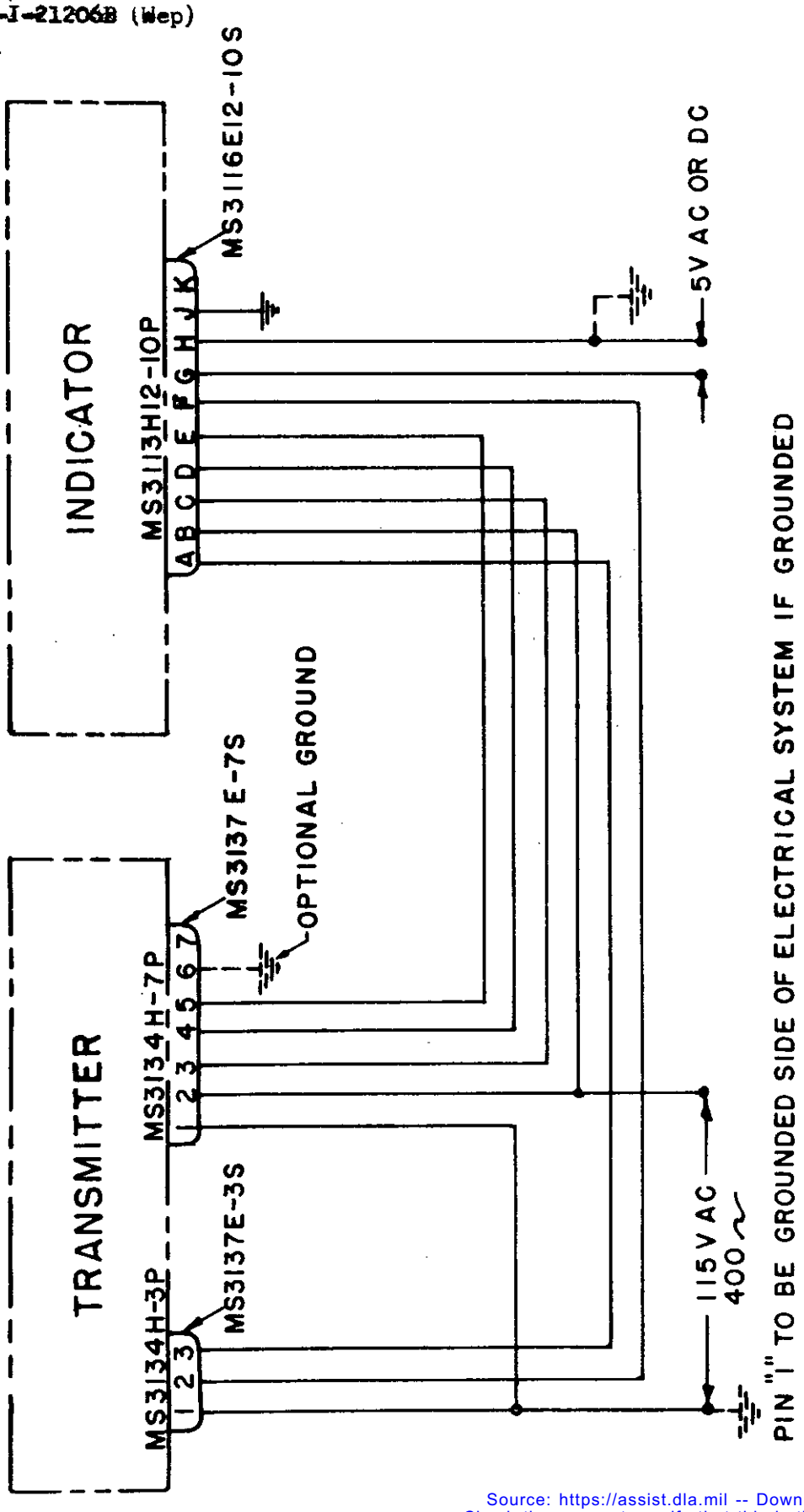


FIG. 2. EXTERNAL WIRING DIAGRAM

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3.5.2.1 Electrical zero reference mark. - The dial shall be provided with an electrical zero reference mark equidistant between the first and last graduations of the dial and shall be located on the outer edge of the dial so that it is visible through the window glass. The electrical zero reference mark shall be finished in durable dull white and shall be 1/32 inch long and 0.010 inch wide.

3.5.2.1.1 Electrical zero. - Electrical zero shall be defined as synchro zero as specified in Specification MIL-S-20708.

3.5.2.2 Other markings. - The type designation EFU-7/A shall be permanently and legibly marked on the dial in 3/64 inch letters. This marking and all other markings not otherwise specified shall be finished in a lusterless black color.

3.6 Pointer. - The pointer shall conform to Standard MS33585-8 except that the pointer shall be suitable for integral lighting. Portions of the pointer which are shown as shaded on Standard MS33585 shall be finished in a lusterless white color. Portions which are shown as unshaded on Standard MS33585 shall be finished in a lusterless black color.

3.7 Visibility of pointer and dial. - The pointer, numerals, at least 1/16 inch of the shortest graduations, and all other specified markings on the dial with the exception of the electrical zero reference mark shall be visible from any point within the frustum of a cone whose sides make an angle of 15° with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.8 Illumination of pointer and dial. - The indicator shall be integrally lighted in accordance with the requirements of Specification MIL-L-25467.

3.9 Window glass. - The quality of the window glass shall be in accordance with Specification DD-G-451, Type II, Quality AA.

3.9.1 Distance from dial to window glass. - The distance between the dial and the inside of the window glass shall be held to a minimum and shall not exceed 0.188 inch.

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3.10 Case. - The outline dimensions of the case shall conform to Standard MS33639 with a nominal diameter of 2 inches. The dimension indicated as "L" shall not exceed 6.125 inches. The body of the case shall be made of lightweight nonmagnetic metal, uniform in texture and shall have a smooth surface. The case shall be finished with a durable, lusterless, black finish.

3.10.1 Hermetic sealing. - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the case reclosed and resealed, at least three times. This shall be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the indicator may be subjected in normal use in military aircraft.

3.10.2 Filling medium. - The filling medium shall be at least 98 percent purity, free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere or 1/2 atmosphere as required. Where practical, the 100 percent helium filling medium shall be utilized.

3.11 Performance. - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4.

3.12 Weight. - The weight of the complete indicator shall not exceed 1.5 pounds.

3.13 Finish. - Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.13.1 Magnesium alloy parts. - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment shall be used. The treatment shall either conform to Specification MIL-C-13335 or shall be approved by the Bureau of Naval Weapons.

3.13.2 Aluminum alloy parts. - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to

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Specification MIL-A-8625 except that the dial, if of aluminum alloy, small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with chemical film in accordance with Specification MIL-C-5541.

3.13.3 Steel parts. - Steel parts shall be cadmium plated, where practicable, in accordance with Specification QQ-P-416, Type II or III, and of a class that is adequate to achieve the degree of protection required.

3.14 Soldering. - Soldering shall be performed in accordance with Specification MIL-S-6872.

3.15 Color. - Lusterless black color shall conform to Standard FED-STD-595 for color number 37038. Lusterless white color shall conform to FED-STD-595 for color number 37875.

3.16 Dimensions. - Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the indicator, they shall be held or limited accordingly.

3.17 Installation instructions. - The contractor shall furnish with each indicator, one copy of the installation instructions, with illustrations and diagrams. The instructions shall be printed on 8 1/2 by 11 or 11 by 17 inch paper and inserted in an envelope marked:

IMPORTANT
THIS ENVELOPE
CONTAINS INSTRUCTIONS

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The sampling, inspection, and test procedures shall be in accordance with Specifications MIL-I-7566, MIL-L-25467 and as specified herein.

4.1.1 Tolerances. - Applicable tolerances shall be as specified in Sections 3 and 4 and in Tables II and III.

4.1.2 Caution.- The transmitter motor load must be connected to the indicator when the indicator is energized to prevent damage to the frequency control circuit.

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Table II

INDICATOR FUEL FLOW TOLERANCE (±PPH)

Test Points PPH	Room Temperature 25° ±2°C	High Temperature 70° ±2°C	Low Temperature -55° ±2°C
300	30	40	40
500	20	30	30
1500	20	30	30
2500	20	30	30
3500	15	25	25
4500	20	30	30
6000	170	250	250
9000	170	250	250
12000	170	250	250
15000	170	250	250

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Table III

INDICATOR TEST TOLERANCES

Requirements	Tolerance ±PPH 300-4900	5000-15000
Position Error at 3500 PPH	15	----
Voltage and frequency variation effects tested on a system basis	±1/2% of point above 1500 PPH except break point and 7.5 PPH below 1500 PPH	
Friction		
Room temperature	20	165
High temperature	20	165
Low temperature	40	250
Vibration		
Pointer oscillation (double amplitude)	25	200
Pointer variation	25	200
Response Time 300-15000 PPH	5 seconds max. 2 seconds min.	
The pointer shall not overshoot the final position more than 1 time		
	Room Temp. Tolerance	
Scale Error After Endurance	± 10	±30

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4.2 Test methods. -

4.3 Individual tests. -

4.3.1 Sealing. - The indicator shall be tested for leaks by means of a mass spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.010 micron cubic foot per hour at a pressure differential of 1 atmosphere or 1/2 atmosphere dependent upon the filling pressure. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 0.10 micron cubic foot per hour. This test shall be conducted in lieu of the sealing test specified in Specification MIL-I-7566.

4.4 Sampling plan A. -

4.4.1 Frequency inverter voltage variation. - A transmitter motor load shall be connected to the indicator and a test voltage of 115V, 400 cps supplied to the indicator. The voltage supplied to the motor of the transmitter shall be measured and the voltage shall not exceed 115V ±10%. The frequency of the voltage shall then be measured with an electronic counter similar to Hewlett Packard Model 522B. The frequency variation shall not exceed 400 cps ±.25 percent.

4.4.2 Low temperature operation. - The indicator shall be placed in a test chamber and properly connected except that no power shall be applied. The instrument shall then be subjected to an ambient temperature of -54°C(-65°F) for a period of 4 hours. During the last 5 minutes of the 4-hour period, power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at -54°C(-65°F) the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Scale error and friction. The tolerances shall be as specified in Table III.

2. Frequency inverter voltage variation (see 4.4.1).

4.4.3 High temperature operation. - The indicator shall be placed in a test chamber and properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of 71°C(160°F) for a period of 4 hours. During the last 5 minutes of the 4-hour period, power shall be applied to the indicator. At the end of the 4-hour

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period and with the temperature maintained at 71°C(160°F), the indicator shall be subjected to and shall meet the requirements specified in the following tests:

1. Scale error and friction. The tolerances shall be as specified in Table III.

2. Frequency inverter voltage variation (see 4.4.1).

4.4.4 Vibration. -

4.4.4.1 Indicator. - This test shall be conducted in accordance with vibration test procedure IV of Specification MIL-E-5272 at an amplitude of 0.009 to 0.011 inch diameter. Test conditions shall be applied to the system to provide an indication of approximately half-scale. Pointer oscillation and variation shall not exceed the tolerances specified in Table III. After vibration the indicator shall be subjected to the scale error test. The scale errors shall not exceed those specified in Table III.

4.4.4.2 Frequency inverter voltage variation. - This test shall be conducted in accordance with vibration test procedure IV of Specification MIL-E-5272 at an amplitude of 0.009 to 0.011 inch diameter. While being vibrated, the indicator shall meet the frequency inverter voltage variation test (see 4.4.1).

4.4.5 Voltage and frequency variation effect. - This test shall be run with the transmitter load on the indicator and tested on a system basis. The applied voltages and frequencies for the test shall be set at the following points:

115V	400 cps
92V	320 cps
138V	480 cps

The voltage and frequency variation effect shall not exceed the tolerance specified in Table III.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - The indicator shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method 1A5, omitting preservation compound, using metal reusable containers (see 6.2).

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5.1.2 Level C. - When this level is required, packaging shall be in accordance with standard commercial practices.

5.2 Packing. -

5.2.1 Levels A and B. - The indicator shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract (see 6.2).

5.2.2 Level C. - The indicator shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers' rules and regulation in effect at the time of shipment.

5.3 Marking. - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary marking. - The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use. - The rate of flow fuel indicator covered by this specification is intended to indicate the rate of flow of fuel in pounds per hour being consumed by an aircraft jet engine when used in conjunction with a TRU-12/A transmitter conforming to Specification MIL-T-21205 or other suitable transmitter.

6.2 Ordering data. - Requisitions, contracts, and orders should state the quantity desired and the levels of packaging (see 5.1) and packing (see 5.2) to be furnished and whether sampling plan "B" is to be omitted.

6.3 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed

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by that date. The attention of the supplier is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Bureau of Naval Weapons, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-9279

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